

Spam Mail Detector Using Machine Learning

1. Project Title

Spam Mail Detector Using Machine Learning and Natural Language Processing

2. Introduction

Email and SMS spam detection is one of the most common applications of Machine Learning and Natural Language Processing (NLP). Spam messages often contain advertisements, fraudulent offers, phishing attempts, or unwanted content that can negatively impact users.

The objective of this project is to develop a machine learning model capable of classifying messages as either **Spam** or **Ham (Non-Spam)**. The project involves text preprocessing, feature extraction using TF-IDF, model training, and performance evaluation using classification metrics.

3. About the Dataset

The project uses the **SMS Spam Collection Dataset** obtained from the UCI Machine Learning Repository.

Dataset Characteristics

- Total Messages: **5,572**
- Classes:
 - Ham (Non-Spam): **4,825**
 - Spam: **747**
- Features:
 - Message Text
- Target Variable:
 - Label (Spam or Ham)

The dataset contains real SMS messages labelled as spam or ham and is widely used for text classification tasks.

4. Exploratory Data Analysis (EDA)

The dataset was initially explored to understand its structure and distribution.

Data Inspection

- Dataset Shape: **(5572, 2)**
- Columns:
 - Label
 - Message
- Missing Values: **None**

Class Distribution

The dataset is slightly imbalanced, with ham messages significantly outnumbering spam messages.

- Ham Messages: 4,825
- Spam Messages: 747

A count plot was created to visualize the distribution of spam and ham messages.

Observations

- Most messages belong to the ham category.
- The dataset contains sufficient examples of spam messages for model training.
- No missing or corrupted records were identified.

5. Methodology

Step 1: Data Loading

The SMS Spam Collection dataset was loaded into a Pandas DataFrame and assigned meaningful column names.

Step 2: Text Preprocessing

To prepare the textual data for machine learning, the following preprocessing techniques were applied:

- Conversion of text to lowercase
- Removal of punctuation marks
- Tokenization of text into words
- Removal of English stopwords

These steps help reduce noise and improve the quality of the textual features.

Step 3: Feature Extraction

Machine learning models cannot process raw text directly. Therefore, the cleaned text was transformed into numerical vectors using:

TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF assigns higher importance to meaningful words while reducing the impact of commonly occurring words.

The resulting feature matrix contained:

- 5,572 messages
- 9,455 text features

Step 4: Train-Test Split

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

Stratified sampling was used to preserve the class distribution in both datasets.

Training Data Shape:

- (4457, 9455)

Testing Data Shape:

- (1115, 9455)

Step 5: Model Training

Two machine learning algorithms were trained and evaluated:

1. Multinomial Naive Bayes

Naive Bayes is widely used for text classification due to its simplicity and effectiveness.

2. Logistic Regression

Logistic Regression is a supervised classification algorithm commonly used for binary classification problems.

6. Model Evaluation and Comparison

The models were evaluated using:

- Accuracy
- Precision
- F1 Score
- Confusion Matrix
- Classification Report

A. Naive Bayes Results

Metric Value

Accuracy 96.14%

Precision 100%

F1 Score 83.14%

Confusion Matrix

Actual / Predicted	Ham	Spam
Ham	966	0
Spam	43	106

Interpretation

- Correctly classified all ham messages.
- Achieved perfect precision.
- Demonstrated strong spam detection capability.
- Highest F1 Score among the tested models.

B. Logistic Regression Results

Metric Value

Accuracy 95.16%

Precision 100%

F1 Score 77.87%

Confusion Matrix

Actual / Predicted	Ham	Spam
Ham	966	0
Spam	54	95

Interpretation

- Also achieved perfect precision.
- Slightly lower recall for spam messages.
- Lower F1 Score compared to Naive Bayes.

Model Comparison

Model	Accuracy	Precision	F1 Score
Naive Bayes	96.14%	100%	83.14%
Logistic Regression	95.16%	100%	77.87%

Best Performing Model

Naive Bayes

Reasons:

- Highest Accuracy (96.14%)
- Highest F1 Score (83.14%)
- Perfect Precision (100%)
- Better spam detection performance

7. Conclusion

This project successfully developed a Spam Mail Detection system using Natural Language Processing and Machine Learning techniques. Text messages were cleaned and transformed into numerical representations using TF-IDF feature extraction. Two classification models, Naive Bayes and Logistic Regression, were trained and evaluated.

The results showed that both models performed exceptionally well; however, the Naive Bayes classifier achieved the best overall performance with an accuracy of 96.14% and an F1 Score of 83.14%. Therefore, Naive Bayes was selected as the final model for spam detection.

The project provided practical experience in text preprocessing, feature engineering, machine learning classification, model evaluation, and Natural Language Processing applications.

Skills Gained

- Text Preprocessing
- Natural Language Processing (NLP)
- TF-IDF Feature Extraction
- Machine Learning Classification
- Model Evaluation
- Spam Detection Systems
- Data Analysis using Python